Battery Operated Sprayer



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Battery operated sprayer

What it is used for

A battery-operated sprayer uses a rechargeable battery and pump to apply water, fertiliser solutions, or approved crop-treatment liquids through a nozzle. Use only chemicals authorised for the intended application and follow the product label, equipment instructions, and required protective measures to prevent exposure, spills, and environmental contamination.

Main parts

- Chemical tank and filling opening
- Rechargeable battery and battery connector
- Electric pump and pressure switch
- Spray hose, trigger valve, and lance
- Nozzle and filter or strainer
- Harness, control switch, and charger

Before you start

1. Confirm the intended application and verify that the liquid is approved for the crop, site, and sprayer materials.
2. Read the product label and safety data; identify the required dilution, application rate, re-entry restrictions, and PPE.
3. Inspect the tank, cap, seals, hose, trigger valve, lance, nozzle, battery, and electrical connectors for leaks or damage.
4. Check that the battery is securely fitted and adequately charged; use only the specified charger and battery type.
5. Ensure the nozzle and filter are clean and appropriate for the liquid and target application.
6. Wear the chemical-specific PPE, such as gloves, protective clothing, eye protection, and respiratory protection where the label requires it; check wind direction before outdoor spraying.

How to use

1. Place the sprayer on a stable surface in a well-ventilated preparation area away from drains and water sources.
2. Confirm that the pump is switched off before filling the tank.
3. Prepare and measure the liquid exactly according to the product label; never mix incompatible chemicals.
4. Fill the tank without exceeding its marked capacity and close the cap securely.

5. Check the hose and nozzle connections and ensure that the trigger valve is closed before starting.
6. Fit the harness correctly and position the sprayer securely on the back or on its approved support.
7. Switch on the pump and test the spray pattern in a designated safe area using the approved procedure.
8. Adjust pressure or nozzle setting only within the manufacturer's permitted range and product-label requirements.
9. Apply the liquid evenly to the intended target, avoiding drift toward people, animals, neighbouring crops, and water bodies.
10. Monitor the tank, hose, battery, and spray output during operation; stop immediately if leakage, clogging, or abnormal pump behaviour occurs.
11. Switch off the pump before clearing a blockage; release pressure and follow the manual before removing the nozzle or filter.
12. After application, switch off the unit, remove PPE safely, and clean the sprayer according to the chemical label and manufacturer instructions.
13. Handle leftover liquid, rinse water, and empty containers according to the product label and local hazardous-waste requirements; recharge and store the battery in a dry, safe location.

⚠ Safety

- Never spray an unidentified chemical or use a product contrary to its label.
- Do not eat, drink, smoke, or touch the face while mixing or spraying chemicals.
- Avoid spraying in strong winds, during unsuitable weather, or where people and animals may be exposed.
- Do not open the tank or remove the nozzle while the system is pressurised.
- Keep the battery, charger, and electrical connections away from water and chemical spills.
- Do not pour chemical residue or rinse water into drains, wells, ponds, or natural waterways.

Care & storage

- Rinse and clean the tank, hose, filter, and nozzle according to the chemical label and equipment instructions.
- Inspect seals, hose connections, straps, and nozzle components regularly for wear or leakage.
- Clean blocked nozzles only with the approved tool; never blow through a nozzle by mouth.
- Charge the battery with the specified charger and avoid storing it in excessive heat or moisture.
- Store the clean, dry sprayer securely and keep chemicals in their original labelled containers.

Common mistakes to avoid

- Using an incorrect dilution or exceeding the tank capacity.
- Spraying without label-required PPE or ignoring wind direction.
- Clearing a blocked nozzle while the pump is running or the system remains pressurised.
- Disposing of leftover chemicals or rinse water in drains or water bodies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.